

MOHD KAIF

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Summary

AI Engineer who has shipped production agentic systems to real users—not just notebooks. Built stateful LLM pipelines, autonomous RAG agents, and low-latency FastAPI services using LangChain, LangGraph, OpenAI, and Anthropic APIs, with end-to-end ownership from design through eval and deployment. Graduating in 2026 (GPA: 8.62/10) from AMU; ready to join immediately and own a production AI system from day one.

Experience

Machine Learning Intern, Acro Technologies India Private Limited Oct 2025 – Mar 2026 (*Noida*)

Project: AI-powered Vendor Management System (VMS) Assistant

- Built and shipped an **autonomous agentic pipeline** (LangChain + LangGraph) converting free-text stakeholder queries into structured hiring actions (Requisition, SOW, RFx)—reducing manual routing time by **~60%** in production with zero human intervention on the happy path.
- Architected an **end-to-end RAG pipeline** with pgvector, **hybrid search** (dense + BM25), and **cross-encoder reranking**—improving retrieval relevance by **20%+**; designed as a **stateful, multi-step reasoning** flow with fallback logic for ambiguous inputs.
- Designed a **configurable embedding pipeline** and schema-selection module, increasing SQL generation accuracy by **~30%** and making the system extensible as new data sources were added.
- Delivered **production-grade API serving** via **async FastAPI** on AWS EC2—**concurrent-safe**, low-latency (under 2 seconds end-to-end), with structured logging and observability for live traffic monitoring.

Project: Resume Parser with Relational Named Entity Recognition

- **Fine-tuned BERT** for domain-specific NER achieving **85–90% entity extraction F1**; built an **eval framework** to measure accuracy and latency trade-offs across model variants before promoting to production.
- Optimised the **async inference pipeline** to cut latency by **~40%**, ensuring real-time response under concurrent API load from live client workflows.
- Deployed as a **typed, production FastAPI service** with Pydantic-validated schemas; owned post-deployment issues end-to-end, resolving 3 integration failures within 24 hours through systematic log tracing.

Machine Learning Analyst Intern, NexGen Nextopia

Aug 2024 – Sep 2024 (*Remote*)

Project: Uniqlo Stock Price Prediction

- Improved model stability by **reducing RMSE by ~18%** through feature engineering and scaling on 3+ years of equity data.
- Built a standardised **evaluation framework** (MAE, RMSE, directional accuracy) enabling fast iteration and cost-latency trade-off analysis across all model variants.

Projects

Personalized Learning Buddy with Persistent Memory

[Code]

- Built a **fully autonomous conversational AI agent** for personalised learning using **LangGraph** stateful pipelines and **LangChain** orchestration—shipped to real users with persistent memory tracking learner state across all sessions.
- Implemented **multi-agent workflows** (explain, step-by-step, practice, assessment) with **LLM-as-judge eval framework**—achieving **87% faithfulness** and **80% semantic similarity** as continuous quality signals.
- Delivered **advanced RAG** with HyDE-style retrieval, reranking, and hybrid search (**80% Recall@K**, **70% MRR**); all LLM API calls (OpenAI, Anthropic) wrapped with retry logic and **streaming-ready** response handling.
- Containerised with **Docker**; wrote **end-to-end test suites** covering retrieval, generation, and memory modules—zero context loss across 100% of multi-turn sessions in production.

Blood Pressure Prediction using PPG Signals

[Code]

- Achieved **SBP MAE of 3.52 mmHg** and **DBP MAE of 1.96 mmHg** using a fine-tuned ResNet1D on MIMIC-III; processed **30GB+** of physiological time-series data through a custom normalisation and segmentation pipeline.

Live Parking Spot Detection System

[Code]

- Delivered **98% accuracy** on images and **96% on live video**; optimised inference pipeline (~35% latency reduction) for real-time occupancy detection at production frame rates.

Education

Bachelor of Technology in Artificial Intelligence

2022 – 2026

Zakir Husain College of Engineering & Technology, Aligarh Muslim University

GPA: **8.62 / 10.0**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Machine Learning, Computer Vision

Skills

Agentic Frameworks:	LangGraph, LangChain, CrewAI (familiar), autonomous agents, stateful multi-step pipelines
LLM APIs:	OpenAI, Anthropic (Claude), Gemini, Llama; prompt engineering, streaming, function calling
RAG & Vector DBs:	RAG pipelines, HyDE, hybrid search, reranking, pgvector, FAISS, ChromaDB, embedding pipelines
Eval & Testing:	LLM-as-judge, faithfulness/similarity scoring, adversarial testing, end-to-end test suites
Backend & APIs:	Python (async, typed), FastAPI, Pydantic, REST APIs, PostgreSQL, MongoDB, concurrent-safe serving
MLOps & Infra:	Docker, AWS EC2, Git, GitHub, Linux, structured logging, production deployment
Fine-tuning:	BERT, ResNet1D, HuggingFace Transformers, model evaluation (accuracy, latency, cost)